

# STATISTICAL ANALYSIS OF LIFE EXPECTANCY AND ECONOMIC INDICATORS

## 1. Introduction

This report examines the connection between economic and health-related factors using a global dataset. The analysis focuses on variables such as GDP, life expectancy, and education levels across different countries. The main objective is to explore patterns, trends, and relationships within the data through statistical analysis and visualisation techniques.

## 2. Relationship Between GDP and Life Expectancy

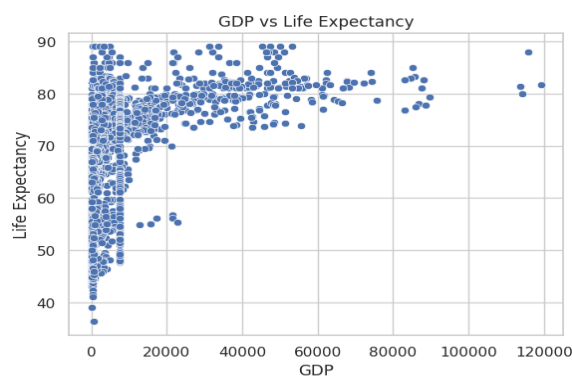


Figure 1. GDP vs Life Expectancy

The scatter plot demonstrates a positive relationship between GDP and life expectancy, showing that countries with stronger economies generally experience better health outcomes. This suggests that economic development plays an important role in improving overall quality of life.

The statistical analysis supports this trend. The average life expectancy is approximately 69 years, while GDP displays high variance, indicating substantial economic differences between countries. In addition, the strong positive skewness of GDP suggests that only a small number of countries possess extremely high economic output. These findings highlight the existence of global economic inequality and its influence on health conditions.

## 3. Comparison of GDP Across Countries

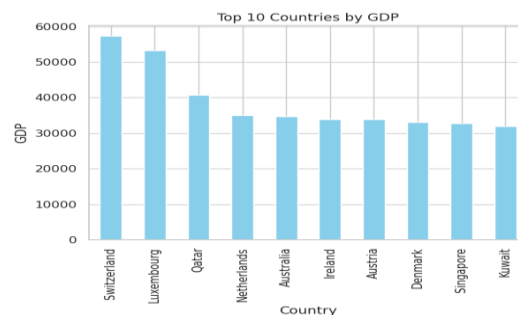


Figure 2. Top 10 Countries by GDP

The bar chart illustrates considerable variation in GDP among the top 10 countries. While a few countries dominate in terms of economic output, others display relatively lower GDP values, reflecting differences in global economic performance.

This is supported by the high variance observed in GDP values. The distribution is also positively skewed, indicating that only a small number of countries contribute significantly to global economic output. This imbalance highlights global economic inequality and may influence differences in living standards and health conditions across countries. These results are consistent with the trend observed in the scatter plot analysis.

4. Correlation Between Key Variables

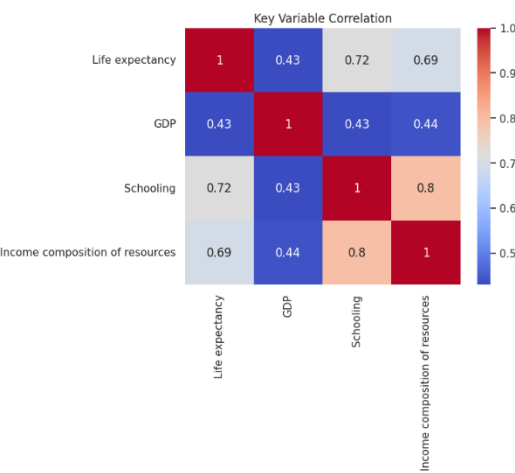


Figure 3. Key Variable Correlation Heatmap

The heatmap shows the relationship between GDP, schooling, income composition, and life expectancy. Strong positive correlations can be seen between economic and educational factors and life expectancy, indicating that countries with better economic conditions and education systems generally achieve improved health outcomes.

These results support the trends observed in the scatter plot analysis and emphasise the importance of socioeconomic development in influencing life expectancy across countries.

5. Conclusion

In conclusion, the analysis demonstrates a strong relationship between economic and health indicators. Countries with higher GDP and better education systems generally tend to have higher life expectancy levels. The statistical findings also highlight global inequality, particularly in terms of GDP and population distribution. Overall, the analysis suggests that economic growth and social development play an important role in improving health outcomes across different countries.